Nate(Xiance) Xu

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EDUCATION

Northwestern University 09/2026 - 06/2028 (Expected)
Master of Science in Electrical Engineering

University of California, Berkeley 01/2025 - 05/2025
GLOBE Exchange Program - Digital Design (EECS 151), Analog IC Design (EE 140), Semiconductor Devices (EE 130)

Xi'an University of Technology 09/2022 - 06/2026
B.Eng. in Electronic Science and Technology | GPA: 3.6/4.0

TECHNICAL SKILLS

HDL & Programming: Verilog, SystemVerilog, C/C++, Python
RTL / FPGA: FPGA Prototyping, AXI4-Lite, AXI-DMA, CDC, BRAM/FIFO, Memory-Mapped I/O, RISC-V, Zynq SoC
EDA / IC Design: Xilinx Vivado, Cadence Virtuoso, Mentor Graphics, DRC/LVS/PEX, Analog/Mixed-Signal IC Design

ENGINEERING EXPERIENCE

Xi'an Institute of Optics and Precision Mechanics, CAS | Research Intern 06/2025 - 09/2025

- Developed and verified a **16-bit ADC sampling interface** in Verilog; resolved clock-domain-crossing instability by redesigning two-stage synchronizers and handshake logic.
- Designed a DDR-less Zynq PL-PS data path using **AXI4-Lite control, BRAM/FIFO double buffering**, Verilog handshake logic, and a memory-mapped PS driver, enabling stable eight-channel acquisition at **50 MHz**.
- Optimized burst length and sampling windows to achieve **145 MB/s measured throughput**; the architecture was adopted as the lab's standard interface for subsequent high-speed acquisition projects.

Xidian University | Research Assistant, Institute of Embedded Computing Technology 07/2024 - 08/2024

- Deployed a MobileNetV2 four-class image classifier to a **Zynq-7000 SoC**, integrating AXI-DMA and PL-side buffering for **12 FPS** inference at 320x240; trained on 2,000 images to **82% test accuracy**.

HARDWARE PROJECTS

Five-Stage RISC-V Processor | EECS 151, UC Berkeley 03/2025 - 05/2025

- Led the design and FPGA implementation of a **32-bit five-stage RISC-V CPU** with CSRs and hardware interrupt handling.
- Achieved **95 MHz Fmax** and **1.24 CPI** with a non-forwarding pipeline, using stall/NOP insertion to simplify hazard logic.
- Reduced LUT use by **22% (1,293 to 1,008)** and saved **34 BRAM blocks**; debugged pipeline hazards and latch inference with assertions and timing reports, and verified through RTL simulation, ISA validation, and FPGA prototyping.

Mixed-Signal IoT IC Design & Tape-Out | EE 140, UC Berkeley 04/2025 - 05/2025

- Led the design and tape-out of a mixed-signal IoT sensor IC integrating an **8-bit SAR ADC, PGA, BGT, analog MUX, and LDO**, supporting **1.6-3.2 V** operation and **0-70 °C**.
- Designed folded-cascode and two-stage amplifiers in **Cadence Virtuoso**; completed **DRC/LVS** and parasitic extraction (**PEX**), and verified a **StrongARM latch comparator** in the SAR ADC path across **PVT corners** post-layout.
- Built AREG/DREG LDOs regulating 1.6-3.2 V to **1.8 V** and achieved **3.7 mV** BGT reference error.

EXPERIENCE & HONORS

CloudSublet | Founder & Engineering Lead 05/2026 - present

- Built a student-housing platform for four university communities on WeChat Mini Program; led an **80-user beta**.

Saviu | Product & Full-Stack Developer | HackUTD Spacetime Track Honorable Mention (5th of 144) 11/2026

- Built with one teammate a real-time, three-portal disaster-response system on 50,000+ simulated households.

Honors: College-Level One-Star Scholarship (2025); University Outstanding Student Scholarship (2023); University Advanced Individual in Scientific and Technological Innovation (2023).